Kushagra Tiwari

Mobile | Mail | LinkedIn | GitHub | Portfolio

EDUCATION

National Institute of Technology Karnataka (NITK), Surathkal Aug 2023 – May 2027
B.Tech. in Electrical and Electronics Engineering CGPA: 7.14/10

EXPERIENCE

Magi | Software Development Engineer Intern Apr 2026 – Oct 2026
TypeScript, NestJS, Next.js, PostgreSQL, Redis, MCP, OAuth 2.0, JWT, Google Chat API, Slack API *Remote*

- Built the **@Magi Google Chat and Slack apps** (slash commands, @mentions, interactive cards) for briefs, brand audits, knowledge search, and approvals, on a transport-neutral **Concierge** layer so Chat, Slack, Asana, and the web app share one service; verified inbound requests via Google-signed JWTs.
- Owned the platform's **MCP server**: **59 schema-validated tools across 8 domains** with RBAC, secured by **OAuth 2.0** (PKCE, refresh tokens, RFC 7591 dynamic client registration) and **Microsoft Entra ID SSO**.
- Added **room-scoped notification delivery** with retry/backoff and permanent-failure cancellation, and shipped Sanity CMS, Asana, and competitive-intelligence agent integrations; **150+ commits** across backend and frontend, backed by **850+ tests**.

PipesHub | Software Development Engineer Intern Dec 2025 – Apr 2026
TypeScript, Node.js, Electron, Python, WebSockets, AWS S3, Azure, GCP, GitHub Actions *Remote*

- Merged **31 pull requests** into an open-source AI platform (**3.2k+ GitHub stars**); owned the Electron desktop app, its build and packaging pipeline, and a **WebSocket proxy gateway** with path-based access control.
- Engineered crash-safe local file synchronization using file watchers, a **write-ahead journal**, resumable batch cursors, deduplication windows, and poison-batch quarantine, feeding Python ingestion services; covered by **250+ tests**.
- Shipped **5 cloud storage connectors** (AWS S3, GCS, Azure Blob, Azure Files, GitLab) with revision-based change detection, and added a **CI integration-test workflow** that gates every pull request.

Acredge | Software Development Engineer Intern Aug 2025 – Dec 2025
Python, LLM APIs, Vector Search, Recombee, n8n, WhatsApp Business API, Webhooks *Remote*

- Cut property-search retrieval latency by **60%** by parallelizing vector and keyword retrieval, collapsing ranking stages, and removing blocking external API calls.
- Extended the AI property-enquiry assistant to **website and WhatsApp** channels using webhook-driven inbound message processing on the WhatsApp Business API.
- Built personalized property **recommendations** on Recombee by streaming user–item interaction events, and automated workflows across Google Sheets, email, and internal APIs.

LeastAction Labs | Software Development Engineer Intern Apr 2025 – Aug 2025
React, Material UI, Node.js, Docker, AWS EC2, Nginx, Linux, CI/CD *Remote*

- Shipped **3 core product surfaces** of a self-hosted data workflow platform in React: pipeline authoring, embedded editing, and live execution output.
- Unified task orchestration behind a validated **API contract** to trigger and track sandboxed runs, returning workspace state, execution metadata, and job results.
- Deployed the marketplace and core platform to production on **AWS EC2** with containerized Docker builds and a CI/CD pipeline, configuring **Nginx** as the reverse proxy routing traffic to **2 Linux services**.

PROJECTS

Gapi | FastAPI-Inspired Web Framework for Go Jun 2026 – Jul 2026
Go, net/http, Generics, OpenAPI 3.1, Dependency Injection

- Built a **zero-dependency** Go web framework on the standard net/http stack, using **generics** for compile-time type-safe handlers, route groups, middleware, and dependency injection.
- Auto-generated **OpenAPI 3.1** specs and interactive API docs from typed handler definitions, with RFC 9457 problem-details error responses.
- Implemented request binding and validation for path, query, header, cookie, and body parameters with custom validators; **43 tests** including fuzzing and benchmarks.

VaultLens | Local-First AI Memory Engine May 2026
Electron, TypeScript, SQLite, sqlite-vec, RAG, Embeddings, Ollama

- Built a local-first **RAG** engine that indexes desktop files into **SQLite** + **sqlite-vec** for natural-language search using local embedding models via **Ollama**.
- Implemented semantic chunking, metadata extraction, mandatory source citations, and task extraction to keep LLM answers grounded in indexed data.
- Enforced filesystem isolation via path canonicalization, symlink-escape prevention, and credential-file exclusion, with per-query privacy receipts and **73 tests**.

TECHNICAL SKILLS

Languages: Go, TypeScript, JavaScript, Python, SQL, C++, Rust
Backend: Node.js, NestJS, Express.js, FastAPI, REST APIs, WebSockets, gRPC, Celery, BullMQ, OAuth 2.0, SSO, RBAC
AI / LLM: MCP, AI Agents, LLM Tool Calling, RAG, Embeddings, Vector Search, Ollama, LangChain, LangGraph
Data / Messaging: PostgreSQL, Redis, MongoDB, SQLite, Kafka
Cloud / DevOps: AWS (EC2, S3), Azure, GCP, Docker, Kubernetes, Nginx, Linux, CI/CD, GitHub Actions
Frontend / Mobile: React, Next.js, Electron, Material UI, React Native, Flutter, Swift
Concepts: System Design, Distributed Systems, Event-Driven Architecture, TCP/IP, DNS, TLS, Unit/Integration Testing